The long COVID evidence gap in England

The term long COVID, also known as post-COVID-19 condition, was coined in spring, 2020, by individuals with ongoing symptoms following COVID-19 in response to unsatisfactory recognition of this emerging syndrome by health-care practitioners.[1] In September to November, 2020, clinical codes for persistent post-COVID-19 condition and related referrals were introduced and became available for use by health-care practitioners to record details of clinical encounters in electronic health records (EHRs) in England. EHRs, which cover a large proportion of individuals living in England, are increasingly used to help understand the epidemiology of disease alongside the effectiveness and safety of interventions. Many factors influence the completeness of information in EHRs, including help-seeking behaviour of patients and the discretion and data-recording behaviour of practitioners. Longitudinal population-based studies often include participant self-reports of illness; hence, these studies might be subject to reporting and participation biases. Comparing reported illness in studies to recorded illness in the EHRs of the same individuals might be helpful in understanding the epidemiology and clinical recognition of emerging conditions such as long COVID.

We investigated whether individuals with self-reported long COVID between July, 2020, and October, 2021, had received a long COVID diagnosis or referral in the English health-care system after 20–32 months of follow-up (data cutoff: April, 2023), among 6405 participants of ten samples from longitudinal population-based studies that used COVID-19 survey data linked to EHRs in the UK Longitudinal Linkage Collaboration (LLC; appendix pp 1–4).

Self-reported long COVID was defined as reporting 4 or more weeks of ongoing symptoms attributable to COVID-19, per National Institute for Health and Care Excellence 2021 guidelines.[2] Seven of the ten samples from longitudinal population-based studies we used comprised reports of debilitating ongoing symptoms attributable to COVID-19, and the remaining three comprised reports of any ongoing COVID-19 symptoms, as described in our previous research[3] and the appendix (p 4). Long COVID-related health-care interactions were identified from International Classification of Diseases 10th edition and Systematized Nomenclature of Medicine Clinical Terms (known as SNOMED-CT) codes (listed in the appendix [p 5]) from Hospital Episodes Statistics (the national database of English secondary care records) and General Practice Extraction Service Data for Pandemic Planning and Research[4] (the national dataset of English COVID-19-relevant primary care records), respectively, for the period July, 2020, to April, 2023.

Of 6405 participants with data on duration of COVID-19 symptoms and linkage to health records, 896 (14%) self-reported long COVID of any severity in longitudinal population-based study surveys (appendix p 4). Among these 896 participants, just 48 (5·4%; 95% CI 4·1–7·0) were identified as having long COVID-related codes in EHRs, with codes assigned within a mean 5·4 months of symptom duration reporting. When restricting to individuals reporting a history of debilitating long COVID, this proportion was only marginally higher (6·3% [95% CI 4·3–9·2]; 25 of 395), with codes assigned within a mean 5·6 months of symptom duration reporting. In analyses of differences in coding by sociodemographic characteristics (table), likelihood of receiving a long COVID EHR code differed by age tertile, with the likelihood being highest among people of middle age (tertile 2; mean age 45·8 years) and lower in younger (tertile 1; mean age 25·2 years) and older (tertile 3; mean age 63·4 years) participants. Coding likelihood did not differ notably by sex or socioeconomic position. However, participants who reported they were of White ethnicity were more likely to receive a code than individuals of other ethnicities.

Caveats to these results should be considered. There are several reasons why self-reported symptoms might not lead to coding in EHRs. For instance, for self-reported symptoms to be apparent in EHRs they must be perceived by the individual experiencing them to be severe

	Absolute percentage difference between groups	p value
Female sex (reference: male sex)	0·9% (–2·2 to 4·0)	0·59
White ethnicity (reference: other ethnicity)*	6·5% (–0·4 to 13·6)	0·03†
Age group		
Tertile 2 (mean age 45·8 years; reference: tertile 1, mean age 25·2 years)	4·9% (1·0 to 8·8)	0·02
Tertile 2 (mean age 45·8 years; reference: tertile 3, mean age 63·4 years)	3·7% (0·0 to 7·4)	0·05
Index of multiple deprivation		
Tertile 2 (reference: tertile 1, most deprived)	0·8% (–3·2 to 4·7)	0·70
Tertile 3 (least deprived; reference: tertile 1, most deprived)	1·1% (–2·7 to 5·0)	0·57

Data in parentheses are 95% CIs. Index of multiple deprivation is a measure of socioeconomic position. p values were calculated using a two-sample test of proportions *Ethnicity data were only available for 736 (82%) of 896 participants in the sample; all other comparisons used a sample size of 896. †A one-sided statistical test was adopted a priori for the comparison of coding in individuals of White ethnicity versus other ethnicities, because evidence suggests that the proportion of individuals with codes would be higher among White individuals than those of other ethnicities.[5]

Table: **Percentage differences in long COVID coding in electronic health records between sociodemographic groups, among individuals with a history of self-reported long COVID (n≤896)**

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For the **UK Longitudinal Linkage Collaboration website** see https://ukllc.ac.uk/

See Online for appendix

enough to warrant seeking care, where care is seen by the individual as being potentially helpful. Additionally, where help is sought, data recording in clinical encounters is influenced by practitioner recognition of signs and symptoms, beliefs on the importance of these, and appropriate coding in EHRs. This issue has been raised previously, with long COVID coding in EHRs differing by clinical software system used to an extent not plausibly explained by true differences in illness prevalence.[6] Individuals with long COVID presenting to health-care practitioners might have been coded with another condition or remained without a code assigned before or during 2021–22 because specific codes were not available for much of 2020 and use of codes was not universal thereafter (although coding increased after the UK National Health Service [NHS] issued an enhanced service specification to general practitioners in July, 2021). Other issues might have influenced our findings, including limits to EHR coverage (eg, if participants in the longitudinal population-based studies had emigrated from England), the possibility of clinical recording of long COVID existing in free-text records that were inaccessible to us, and selection bias resulting from longitudinal population-based study participation. However, these considerations are unlikely to provide a complete explanation for the substantial discrepancy we observed between perceived and reported long COVID and long COVID coding apparent in EHRs.

For more on the **CONVALESCENCE Study Collaborative** see https://www.ucl.ac.uk/covid-19-longitudinal-health-wellbeing/convalescence-study-collaborative

In conclusion, we found a striking discrepancy between occurrence of long COVID as perceived and reported by participants in longitudinal population-based studies and evidence of long COVID recorded in their EHRs. This finding might reflect substantial unmet clinical need, in keeping with reports from individuals with long COVID of difficulties accessing health care and suboptimal recognition of and response to their illness when they do.[5] Our data might also suggest that unmet needs could be higher among individuals of non-White ethnicity,[7] although confirming this finding would require further investigation in larger quantitative and qualitative analyses. Moreover, our results indicate potential shortcomings of epidemiological research on emerging conditions, such as long COVID, using EHR data or longitudinal population-based study data alone, which might not be recognised or acknowledged in published research studies. These resources each have distinct strengths and weaknesses for identifying long COVID in populations, and our research illustrates the value of triangulation between these resources when these data are available for the same individuals and made accessible efficiently and in compliance with good data sharing practices through research resources such as the UK LLC.

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